

1 Evidence

Pairwise outcomes and observation counts

$$w_{ab}, w_{ba}, m_{ab}$$

Optional context labels c and target mixture q
Keep ties and the observation mask.

2 Statistical target

Estimate reciprocal pair probabilities

$$\hat{P}_{ab} + \hat{P}_{ba} = 1$$

Counts or a contextual pairwise model

With labeled contexts:

$$\hat{P}(q) = \sum_c q_c \hat{P}^{(c)}$$

3 Solution operators

STE-LSE: smooth plug-in path

$$D_\tau = \sigma((\hat{P} - 1/2)/\tau)$$

zero diagonal

Top Cycle: soft reachability
Paths up to $K \rightarrow t$

Uncovered Set: soft covering
direct win + no witness $\rightarrow u$

4 Report

Membership scores
 $t(a), u(a)$

Declare size k or threshold h
Report coverage, DG, and bootstrap sensitivity

Scores $\neq$ probabilities

STE-Post: separate uncertainty path

Specify edge posterior $\Pi \rightarrow$ sample hard tournaments
 $\rightarrow$ compute hard TC/UC $\rightarrow$ average membership indicators

Inclusion frequencies
 $\bar{t}_S(a), \bar{u}_S(a)$
conditional on Π

Solid: plug-in flow
Dashed: sampling branch

Conceptual workflow. Context reweighting needs labeled support; no branch corrects unobserved judge bias.